Assignment1_Extra solution

1. DFF

- **Design**

  The DFF design has missing begin end that causes a bug in the results which is when enable signal is zero the output q signal still updates with the changes in the input d

```rtl
module dff(clk, rst, d, q, en);
parameter USE_EN = 1;
input clk, rst, d, en;
output reg q;

always @(posedge clk) begin
    if (rst)
        q <= 0;
    else
        if(USE_EN)
            if (en)
                q <= d;
        else
            q <= d;
end

endmodule
```

design before justifications

```rtl
module dff(clk, rst, d, q, en);
parameter USE_EN = 1;
input clk, rst, d, en;
output reg q;

always @(posedge clk) begin
    if (rst)
        q <= 0;
    else
        if(USE_EN) begin
            if (en) q <= d;
        end
        else q <= d;
end

endmodule
```

design after justifications

- **Testbench**

  The two testbenches are the same except the USE_EN parameter value.

```rtl
module dff_t1_tb ();
  //? local parameters
  localparam USE_EN = 1;
  localparam CLK_CYCLE = 2;
  //? Interface signals
  logic clk, rst, d, en, q;
  //? temp signals
  logic q_temp;
  //? Variables
  int error_count, correct_count;
  //? DUT instantiation
  dff #(
      .USE_EN(USE_EN)
  ) DUT_dff1 (
      .clk(clk),
      .rst(rst),
      .d  (d),
      .en (en),
      .q  (q)
  );
  //? CLK generation
  always #(CLK_CYCLE / 2) clk = ~clk;
  //? Driver block
  initial begin
    clk = 0;
    rst = 0;
    d   = 1;
    en  = 0;
    #CLK_CYCLE;
    assert_reset();
    if (USE_EN == 1) begin
      en = 0;
      q_temp = q;
      for (int i = 0; i < 10; i++) begin
        d = $random;
        check_result(q_temp);
      end
      en = 1;
      for (int i = 0; i < 10; i++) begin
        d = $random;
        check_result(d);
      end
      en = 0;
      q_temp = q;
      for (int i = 0; i < 10; i++) begin
        d = $random;
        check_result(q_temp);
      end
    end else begin
      for (int i = 0; i < 10; i++) begin
        d = $random;
        check_result(d);
      end
```

```
    end
  end
  $display("error_counts = %0d, correct_counts = %0d", error_count, correct_count);
  $stop;
end

//? TASKS
task assert_reset();
  rst = 1;
  check_result(0);
  #(5 * CLK_CYCLE);
  rst = 0;
endtask

task check_result(input q_exp);
  #CLK_CYCLE;
  if (q_exp == q) correct_count++;
  else begin
    error_count++;
    $display("xxxx ERROR: the actual output result is not matching the expected result xxxx");
    $display("q=%0b ,q_exp=%0b, rst=%0b, en=%0b", q, q_exp, rst, en);
  end
endtask

endmodule
```

- **DO file**

```
vlib work
vlog dff.v dff_t1_tb.sv dff_t2_tb.sv  +cover -covercells
vsim -voptargs=+acc work.dff_t1_tb -cover
coverage save dff_t1.ucdb -onexit -du dff
run -all
quit -sim
vcover report dff_t1.ucdb -details -annotate -all -output cover_rpt_t1.txt

vsim -voptargs=+acc work.dff_t2_tb -cover
coverage save dff_t2.ucdb -onexit -du dff
run -all
quit -sim
vcover report dff_t2.ucdb -details -annotate -all -output cover_rpt_t2.txt

vcover merge dff_merged.ucdb dff_t1.ucdb dff_t2.ucdb -du dff
vcover report dff_merged.ucdb -details -annotate -all -output cover_rpt_merged.txt
```

- **Coverage report**

❖ **Branch coverage (merged)**

```
Coverage Report by instance with details

================================================================================
=== Instance: /\work.dff
=== Design Unit: work.dff
================================================================================
Branch Coverage:
    Enabled Coverage              Bins      Hits    Misses  Coverage
    ----------------              ----      ----    ------  --------
    Branches                         4         4         0   100.00%

================================Branch Details================================

Branch Coverage for instance /\work.dff

    Line         Item                      Count     Source
    ----         ----                      -----     ------
  File dff.v
------------------------------------IF Branch------------------------------------
    7                                        32     Count coming in to IF
    7               1                         4        if (rst)

    11              1                        10             if (en) q <= d;

                                             11     All False Count
    9               1                         7        else

Branch totals: 4 hits of 4 branches = 100.00%
```

❖ **Statement coverage (merged)**

```
Statement Coverage:
    Enabled Coverage              Bins      Hits    Misses  Coverage
    ----------------              ----      ----    ------  --------
    Statements                       4         4         0   100.00%

================================Statement Details================================

Statement Coverage for instance /\work.dff  --

    Line         Item                      Count     Source
    ----         ----                      -----     ------
```

❖ Toggle coverage (merged)

```
Toggle Coverage:
    Enabled Coverage              Bins      Hits    Misses  Coverage
    ----------------              ----      ----    ------  --------
    Toggles                         10        10         0   100.00%

================================Toggle Details================================

Toggle Coverage for instance /\work.dff  --

                                                  Node      1H->0L      0L->1H  "Coverage"
                                                  ----------------------------------------
                                                   clk           2           2      100.00
                                                     d           2           2      100.00
                                                    en           1           1      100.00
                                                     q           2           2      100.00
                                                   rst           2           2      100.00

Total Node Count     =          5
Toggled Node Count   =          5
Untoggled Node Count =          0

Toggle Coverage      =     100.00% (10 of 10 bins)


Total Coverage By Instance (filtered view): 100.00%
```